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To whom it may concern

This is to certify that Mr. Ananyo Sen, an M.Sc. student in Data Science and Artificial Intelligence at the Ramakrishna Mission Vivekananda Educational & Research Institute, has successfully completed his summer internship at the FLAME School of Computing and Data Sciences, FLAME University, under the supervision of Dr. Chiranjoy Chattopadhyay.
During the internship, Mr. Sen carried out a comprehensive literature-based research project titled "***Multimodal Understanding of Comics and Visual Narratives: A Literature Review and Research Framework***". His work focused on analyzing key technological advancements and challenges in multimodal learning and visual document understanding.

As part of his research, Mr. Sen systematically evaluated state-of-the-art research works covering essential domains including automatic panel extraction, page navigation and reading-order determination, stylized cartoon image processing, optical character recognition (OCR), multimodal emotion recognition, cross-panel reasoning and narrative closure, as well as state-of-the-art vision-language model benchmarks such as ComicsPAP.

He critically synthesized these components, identified major domain challenges, and constructed a robust research framework outlining future experimental methodologies for grounded narrative understanding.

Throughout his internship, Mr. Sen demonstrated high academic standards, strong self-direction, and exceptional analytical capabilities. I wish him all the success in his future research and academic endeavors.

Chiranjoy Chattopadhyay
Associate Professor
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FLAME University, Pune, Maharashtra